

Peer Review

Derivation of the Analytic Solution of the Coupled DL–AL Problem via the Time–Laplace / Resolvent Construction

Referee report

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Recommendation

Major revision. The overall strategy (Laplace transform $\rightarrow$ resolvent $\rightarrow$ pole spectrum) is correct and well motivated, the algebra I checked is sound, and the validation section is commendably honest. However, there is a substantive disconnect between the construction *derived* (Secs. 5–8) and the one actually *validated* (Sec. 9), together with an internal inconsistency in the pole/residue bookkeeping that bears directly on the central “analytic solution” claim. Both are fixable, largely by reframing rather than re-deriving.

1 Summary of the manuscript

The paper studies a coupled diffusive–layer (DL) / advective–layer (AL) system on the unit disk with N evenly spaced microtubules. It (i) Laplace–transforms in time and eliminates the AL density exactly as a linear functional of the DL trace on a microtubule ray; (ii) rewrites the microtubule sources as a Dirac comb so they feed every angular harmonic; (iii) constructs a closed modified–Bessel radial Green’s function; (iv) closes the problem into a Fredholm equation of the second kind whose secular determinant $\det(\mathcal{I} + \mathcal{A}(s)) = 0$ yields the (complex) decay rates; (v) inverts by the Bromwich contour to a modal residue series; and (vi) gives a two–mode truncation and a numerical validation against a finite–difference solver across the (v, w) parameter plane.

2 Strengths

- The motivation for a resolvent (rather than eigenfunction) construction on a non–self–adjoint operator (Sec. 1) is correct and clearly argued.
- The exact AL elimination [Eq. (6)] is clean and correct.
- The Dirac–comb, full–disk reformulation (Sec. 3) is elegant and is the right way to dissolve the apparent “extra boundary condition” / cusp paradox of the single–wedge treatment.
- The radial Green’s function [Eq. (13)] is correct; I verified the Abel constant and the central– δ collapse to the closed form [Eq. (15)].
- The modal–amplitude formula [Eq. (22)] via a biorthogonal projection is correct, and its reduction to the classical Fourier–Bessel coefficient in the uncoupled limit (Sec. 7 sanity check) is convincing.
- The validation (Sec. 9) is transparent and, unusually, reports a regime in which the method fails rather than only its successes.

3 Major concerns

M1. The derived object (Secs. 5–8) is not the object validated (Sec. 9). Sections 5–8 present the analytic solution as the resolvent / secular determinant $\det(\mathcal{I} + \mathcal{A}(s)) = 0$, which is a *nonlinear* eigenvalue problem: $\mathcal{A}$ depends on s through both the Green’s kernel and the AL functional. Section 9, however, states that the modes are obtained by writing the coupled operator in the Bessel basis and *diagonalizing* it — an ordinary, s -independent (generalized) eigenproblem for the generator. These are related but not identical constructions, and the manuscript silently substitutes one for the other while continuing to cite Eqs. (24)–(25) as though they were what is computed. The paper should either (a) prove the equivalence (generator eigenvalues = roots of $\det(\mathcal{I} + \mathcal{A}(s)) = 0$), or, preferably, (b) promote the generator eigenproblem to the primary construction — it is what actually runs and is validated — and present the resolvent/secular determinant as motivation. As written, Sec. 9 cannot be reproduced from Secs. 5–8.

M2. The “subdominant Φ_0 ” argument (Sec. 6) is internally inconsistent and contradicts Sec. 9. In assembling the residue series, the manuscript discards the forced-response residue family at the bare poles $s = -\kappa_{0,j}^2$, describing it as a “diffusive-background residue family ... which we suppress here as subdominant.” But the slowest bare pole, $-\kappa_{0,1}^2 \approx -5.78$, is the *most* dominant (slowest-decaying) contribution, not a subdominant one. Moreover, the dominant decay rate reported throughout Sec. 9 is $\sigma \approx 5.0$ – 5.8 — i.e. precisely the (perturbed) $n = 0$ pole that Sec. 6 suppresses. The text must resolve whether the bare pole is (a) removable in $(\mathcal{I} + \mathcal{A})^{-1}\Phi_0$, (b) shifted by the coupling (as claimed near Eq. (13)), or (c) a genuinely separate contribution; at present all three are asserted at different points. This is the single most important rigor gap, since it concerns which poles constitute “the spectrum.”

M3. The “truncation breakdown” diagnosis is asserted, not demonstrated. For the failing regime $v = 100$, $w = 100$ (predicted $\sigma = 11.34$ vs. numerical 3.40; mass MSE $\sim 10^{-1}$), the paper attributes the disagreement to a too-coarse few-mode truncation together with first-order discretization effects. No convergence evidence is given: it is never shown that increasing the number of radial/angular modes (or refining the grid) reduces the error monotonically. Without such a study the failure could equally reflect a genuine model mismatch. A mode-count and grid-refinement convergence panel for this regime should be added before the “truncation” explanation can be accepted.

M4. The validated equations are not the continuum equations of Secs. 1–2. Section 9 concedes that the analytic operator is supplied with the discrete scheme’s coupling, $a \rightarrow ar\Delta\theta$ — a grid-dependent weight with *no continuum limit*. Consequently Sec. 9 validates the analytic solution of a grid-modified system, not the continuum problem derived in Secs. 1–8. This caveat should be stated at the *opening* of Sec. 9 (and reflected in the abstract), not introduced midway. Relatedly, the phrase “depend *mildly* on the grid” understates the effect: the paper’s own refinement data move σ by roughly 25% ($\approx 4.2 \rightarrow 5.2$) over $16 \rightarrow 48$ rings. “Mildly” should be removed or quantified.

M5. Non-self-adjoint completeness is assumed silently. The modal series [Eq. (24)] presumes simple poles, the absence of Jordan blocks, and completeness of the biorthogonal family $\{\psi_k\}$. For a non-self-adjoint operator none of these is automatic. At minimum the manuscript should add an explicit caveat; ideally it should argue (or cite) why the expansion is complete for this operator.

4 Minor concerns

- m1. **Notation clash.** a denotes the on-rate throughout, yet Eq. (27) writes the angular-modulation amplitude as $a(r)$. Rename the amplitude (e.g. $\alpha(r)$) to avoid collision with the coupling rate.
- m2. **Addition-theorem hand-wave (Eq. (18)).** The remark that the ray-summed kernel “is the image-corrected $\frac{1}{2\pi}K_0(\sqrt{s}|\mathbf{r} - \mathbf{r}'|)$ ” is loose: the sum runs over orders nN (not all n), and the disk Dirichlet correction is not the free-space K_0 . Either derive the image series or delete the parenthetical and retain S as the mode sum.

- m3. **Undefined block structure.** In the 2×2 truncation [Eq. (30)], $\mathcal{S}_{mn}$ is called “the (m, n) angular block of $\mathcal{S}$,” but $\mathcal{S}$ [Eq. (18)] is defined as a single angle-summed scalar kernel with no (m, n) block structure. The projection producing p_{mn}, q_{mn} is asserted rather than derived.
- m4. **Sign of v .** The continuum AL equation uses $+v \partial_r \rho$; the physical (inward) direction and the sign convention for v are never fixed. State them explicitly (the companion solver negates the velocity input).
- m5. **Branch cut.** The dismissal of the $\sqrt{s}$ cut (Sec. 6) is correct for a bounded domain but would be clearer with a one-line note that $\Phi_0, \mathcal{K}_m$ are even in $\sqrt{s}$ and hence meromorphic in s .
- m6. **Title.** Given that the secular roots are only obtained numerically and the validated object is grid-modified, “Derivation of the Analytic Solution” slightly oversells; “A resolvent construction...” or “semi-analytic” would be more accurate.
- m7. **Reproducibility.** The exact `deterministic_verification` invocation (parameters, mode counts, time stamps) that produced Table 1 should be recorded for the notes’ own future use.

5 Questions to the authors

1. Are the generator eigenvalues of Sec. 9 provably the roots of $\det(\mathcal{I} + \mathcal{A}(s)) = 0$ of Sec. 5? If so, the equivalence should be shown; if not, which set constitutes the physical spectrum?
2. Does refining the mode counts $n_{\max}, J_r$ and the grid reduce the $v = 100, w = 100$ error monotonically?
3. Since the dominant root s_k lies near a bare pole $-\kappa_{0,1}^2$, is the kernel $\mathcal{K}_0(\cdot; s_k)$ not nearly singular there? How is this handled numerically in the reconstruction [Eq. (24)]?

6 Assessment

The physical model and the bulk of the algebra are correct, the comb reformulation is a genuine contribution, and the honest validation (including a failing regime) is a real asset. The required revisions are primarily about *coherence*: align the derived construction (Secs. 5–8) with the validated one (Sec. 9) [M1], repair the pole/residue bookkeeping [M2], and foreground the fact that the validated system is the grid-modified one [M4]. The one outstanding *empirical* task is a convergence study for the failing regime [M3]. With these addressed, the manuscript would constitute a sound and useful account of the coupled DL–AL problem.